

Vessel-break sensitivity in a coupled 3D--1D model Completed controlled pilot, 11 September 2026.
All results are dimensionless and concern synthetic geometries.

Break location has a much larger effect than its small missing length alone suggests. For identical gaps removing 2.5% of a straight vessel, inlet-flux reductions are 68.39%, 29.20%, and 7.67% at 20%, 50%, and 80% of its length. A Y trunk break reduces inlet flux by 66.43%; a single daughter break reduces it by about 4.64%. In the larger tree, 32 breaks give a median reduction of 53.21% and a five-seed range of 35.52--96.00%.

MODEL: Unit cube, circular-average coupling, positive vessel inlet $U=1$; zero exterior tissue potential and sealed other vessel endpoints. $K_3=K_1=1$, radius=0.025, $\kappa=0.05$. Axial flux $q=-K_1\pi r^2 dU/ds$. All quantities dimensionless. Retained disconnected pieces still exchange with tissue.

REGIME: Intact peak tissue potential is less than 0.3% of the imposed inlet value, so feedback on the supplied vessel is weak here. The magnitudes depend on these coefficients and boundary conditions. All test geometries are trees; loops can provide alternative supply paths.

TOY RESULTS:

line | gap_at_0.2 | 68.386 | 76.940 | 80.77

line | gap_at_0.5 | 29.200 | 38.206 | 50.00

line | gap_at_0.8 | 7.674 | 12.254 | 19.23

line | gap_width_0.01 | 28.581 | 37.534 | 50.00

line | gap_width_0.04 | 30.461 | 39.569 | 50.00

y | gap_trunk | 66.432 | 74.833 | 85.64

y | gap_upper | 4.644 | 10.895 | 17.07

y | gap_lower | 4.645 | 10.882 | 17.07

y | gap_both_daughters | 10.527 | 16.151 | 34.66

Columns: geometry, case, inlet flux reduction (%), tissue L2 error (%), detached length (%).

RANDOM BREAKS:

0 breaks: median flux loss 0.000%; range -0.000--0.000%; median tissue error 0.000%.

1 breaks: median flux loss 0.030%; range 0.030--5.391%; median tissue error 0.486%.

2 breaks: median flux loss 0.233%; range 0.060--5.423%; median tissue error 2.297%.

4 breaks: median flux loss 1.535%; range 0.123--11.572%; median tissue error 7.179%.

8 breaks: median flux loss 11.421%; range 5.765--37.576%; median tissue error 21.560%.

16 breaks: median flux loss 18.076%; range 8.213--96.000%; median tissue error 29.382%.

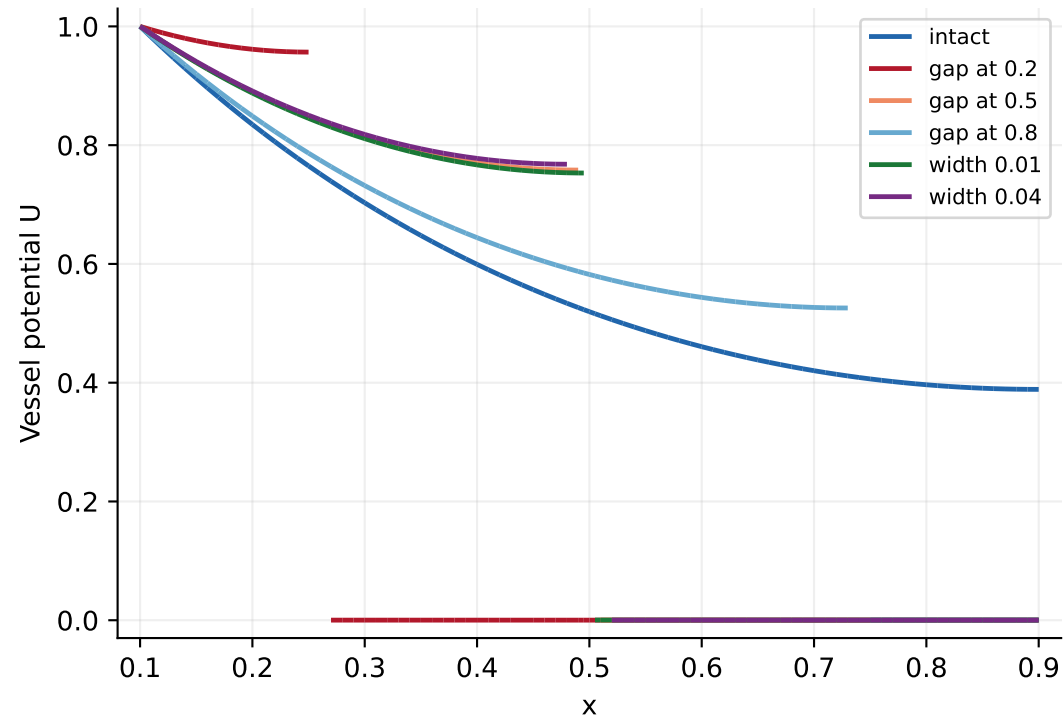
32 breaks: median flux loss 53.211%; range 35.522--96.000%; median tissue error 58.903%.

VALIDATION: Maximum conservation error 8.051e-13; serial/MPI relative difference 1.785e-13; analytic flux order approximately 2.

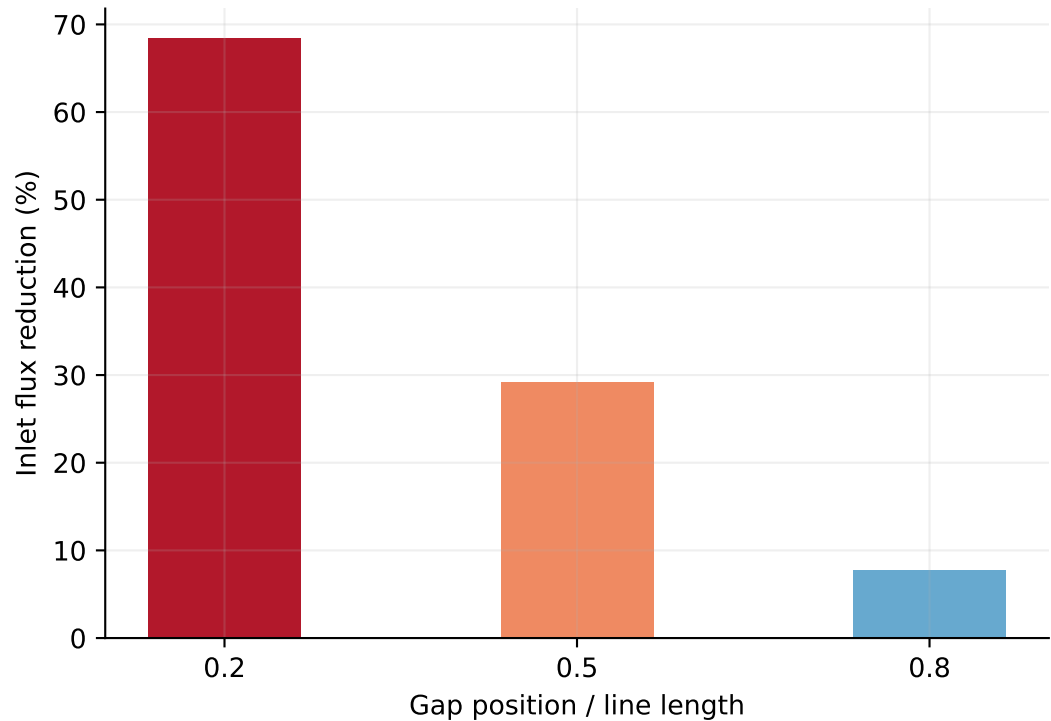
SCOPE: 42 primary synthetic cases. Five random seeds. This is a controlled dimensionless pilot, not a fitted data model. Read REPORT.html or REPORT.md for methods, refinement tables, complete qualifications, and file instructions.

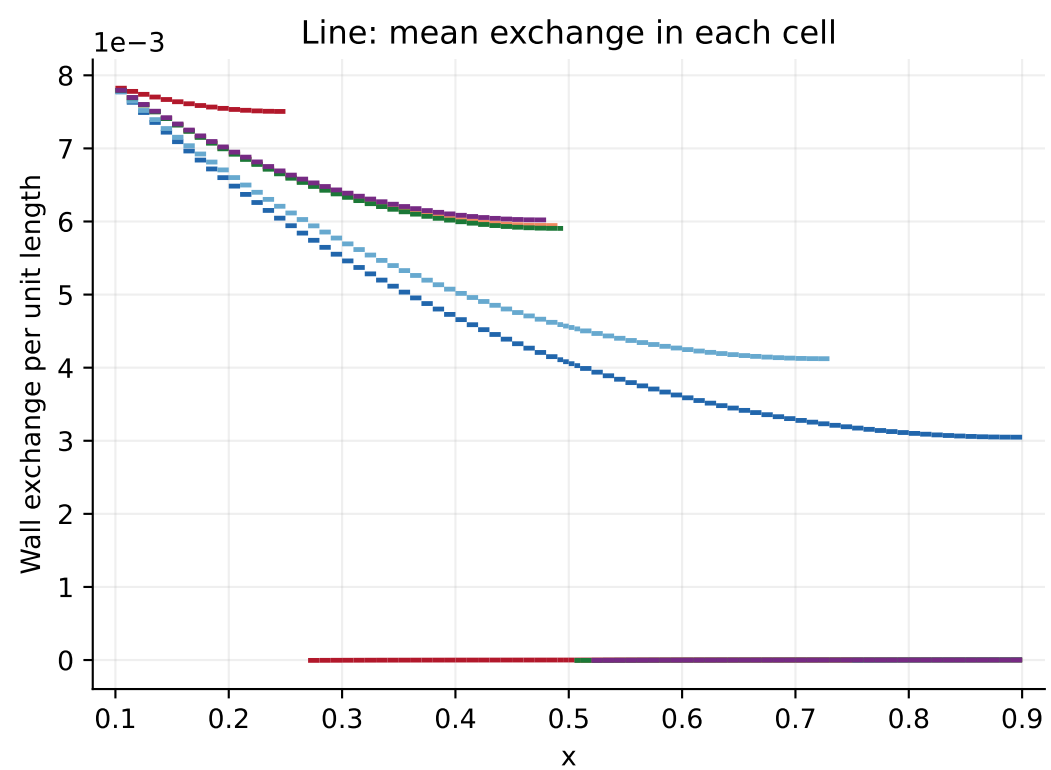
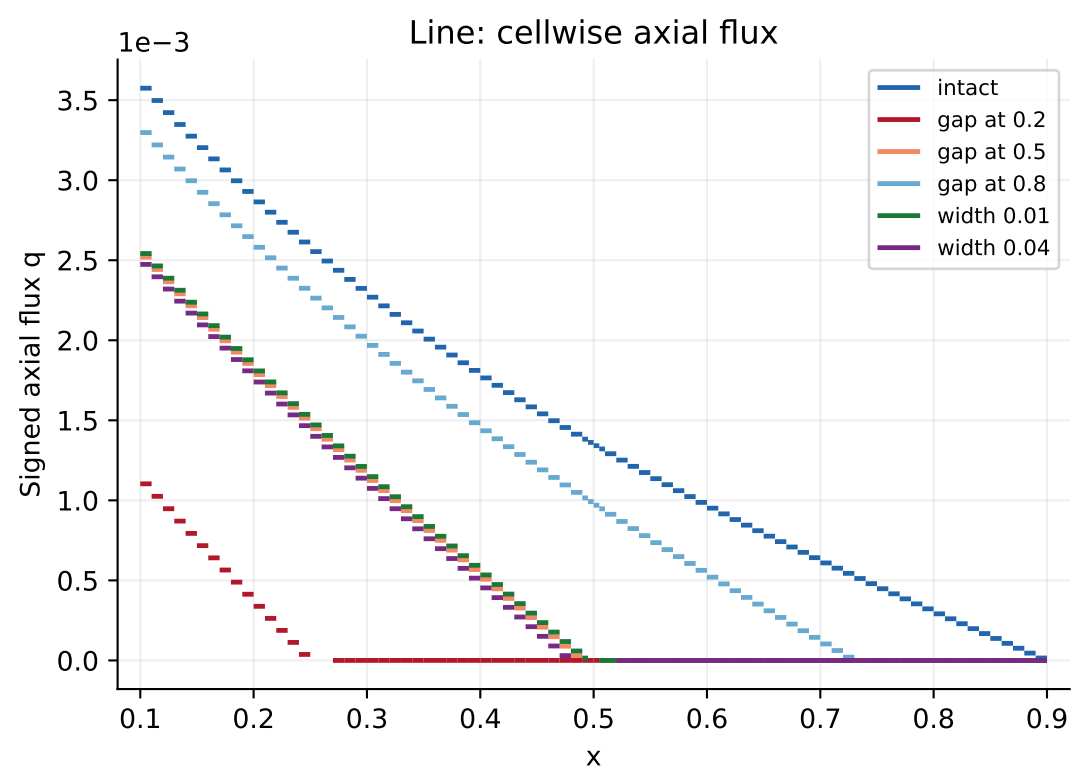
REFERENCE: Laurino and Zunino (2019), DOI 10.1051/m2an/2019042.

Line: pressure on retained segments

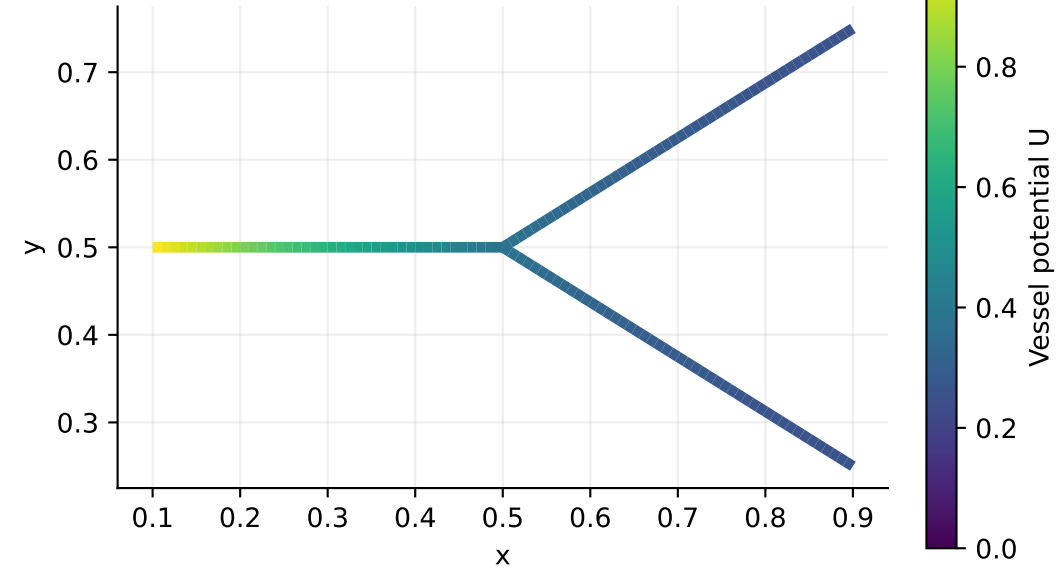


Identical 0.02-long gaps

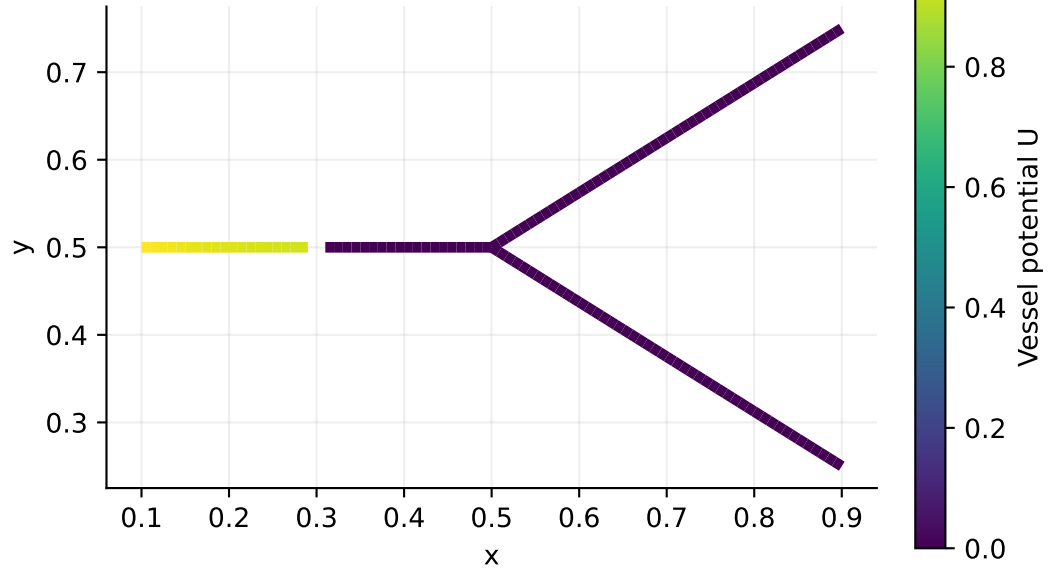




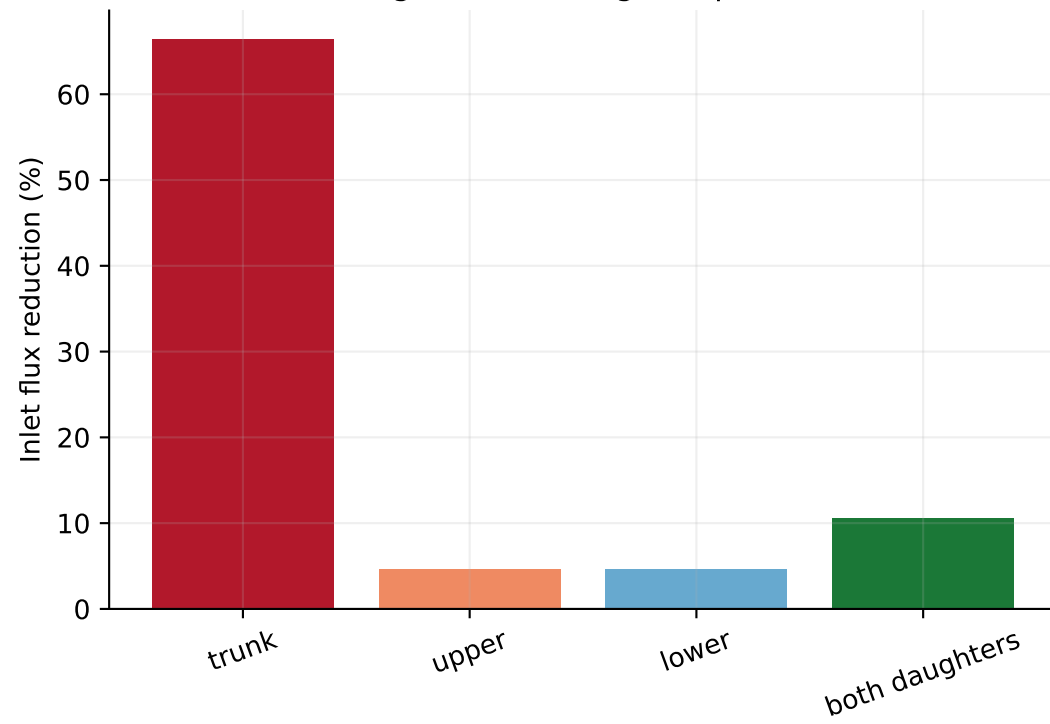
Intact Y



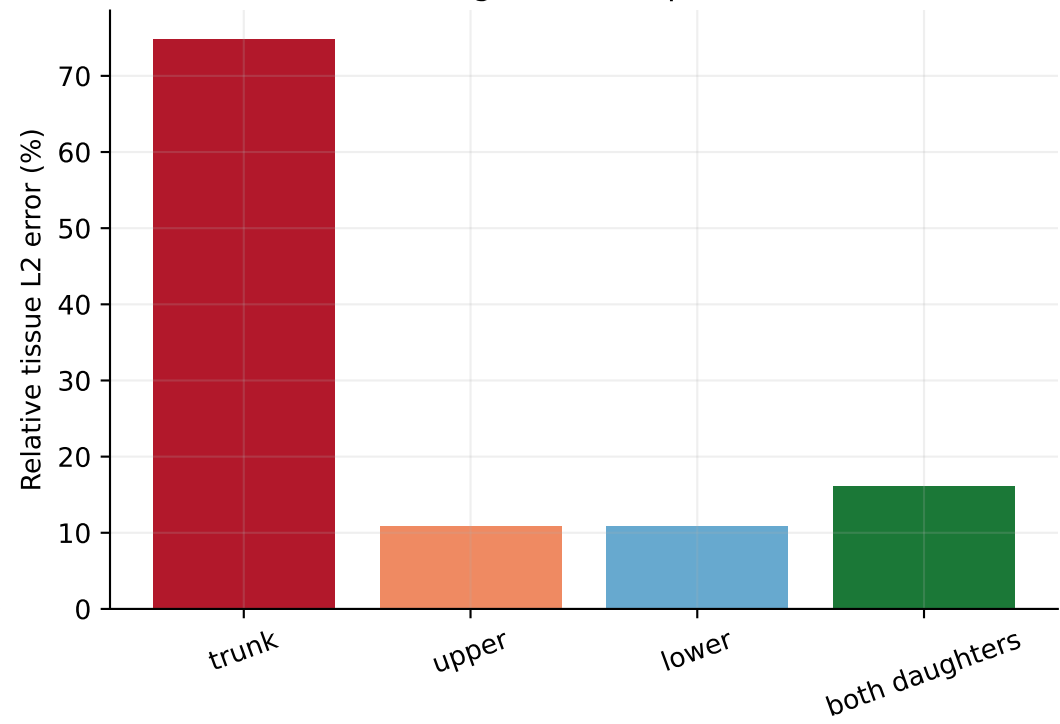
Y with trunk gap

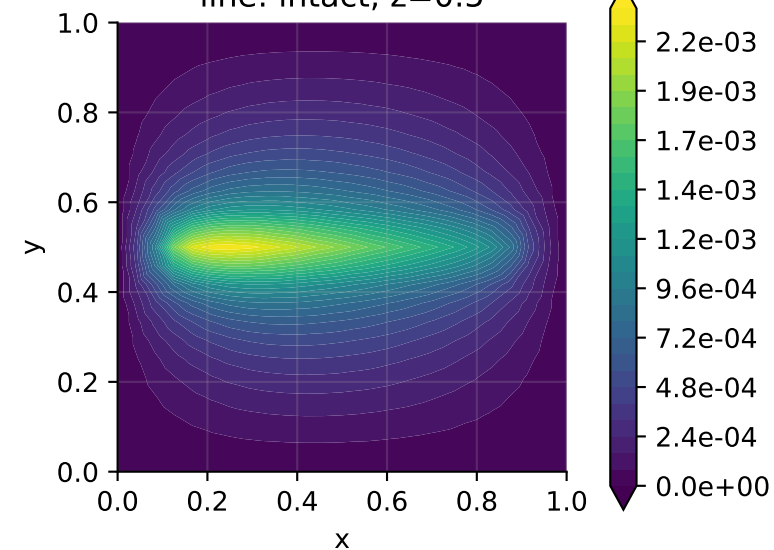
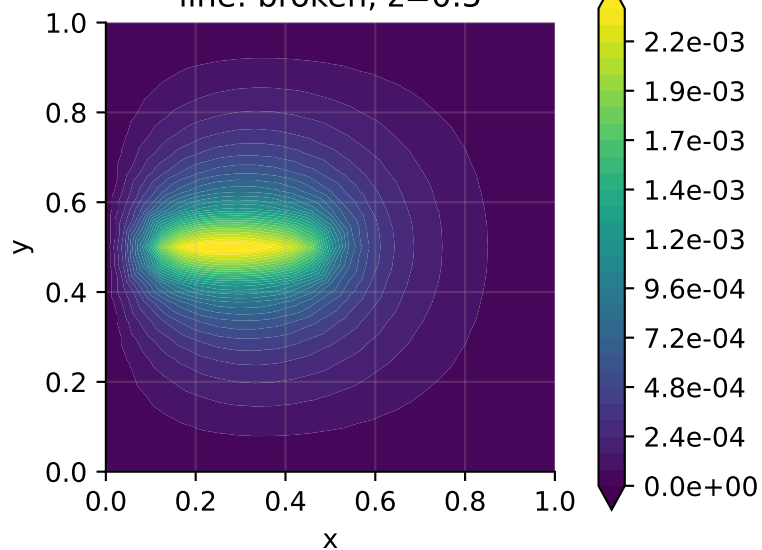
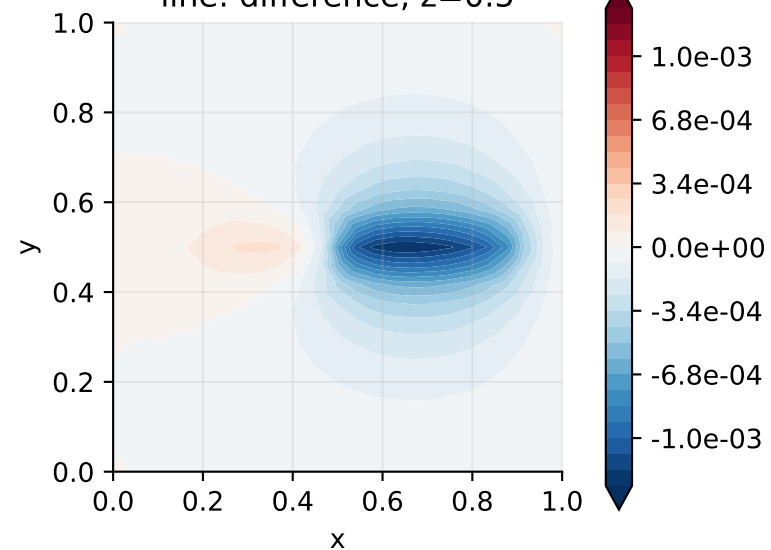
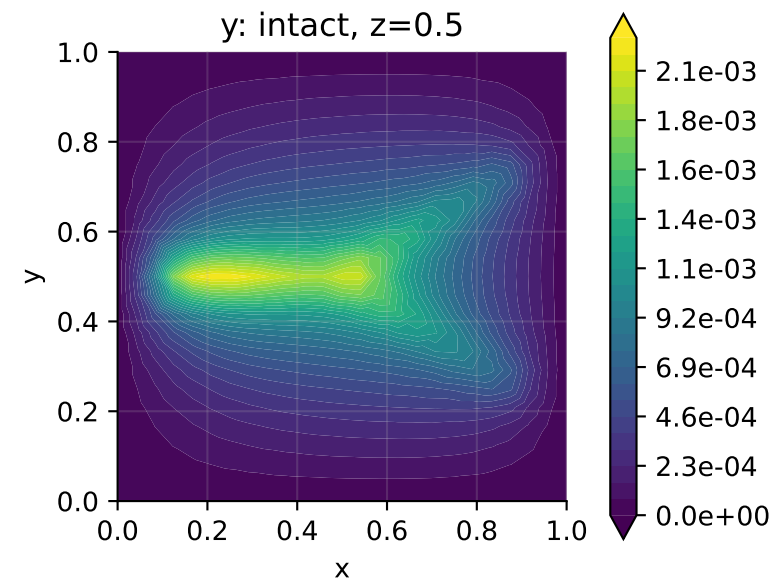
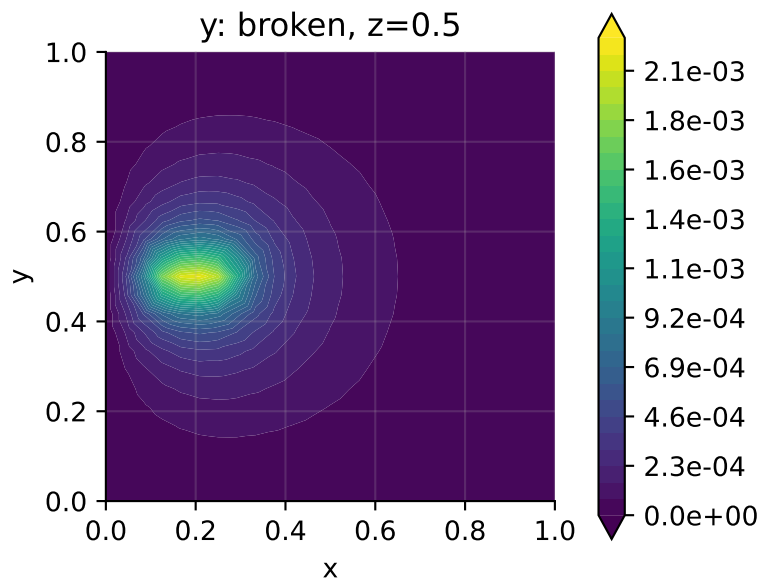
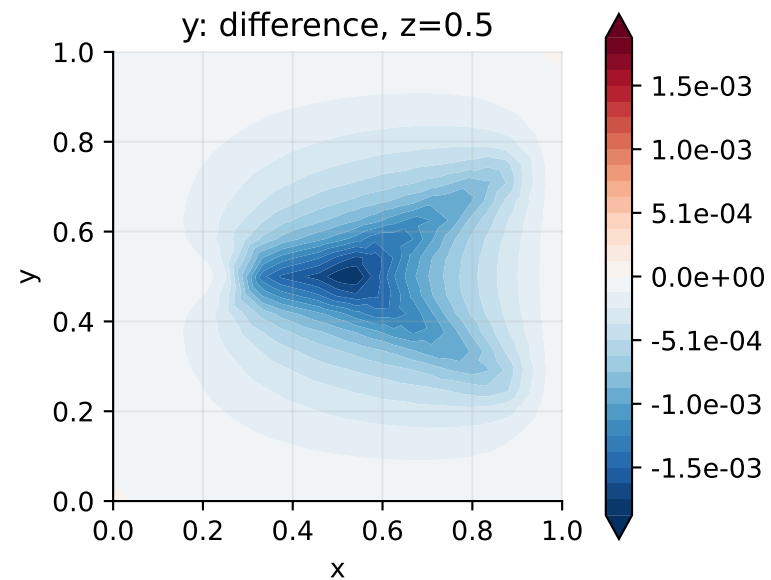


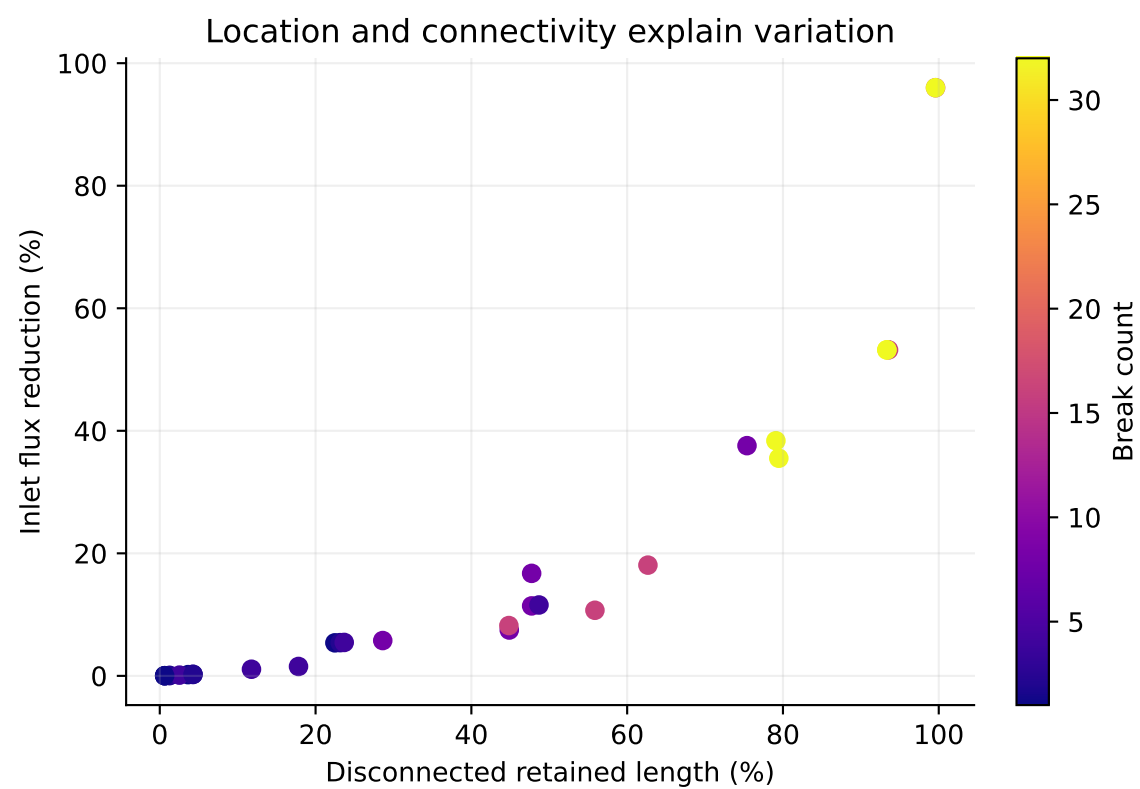
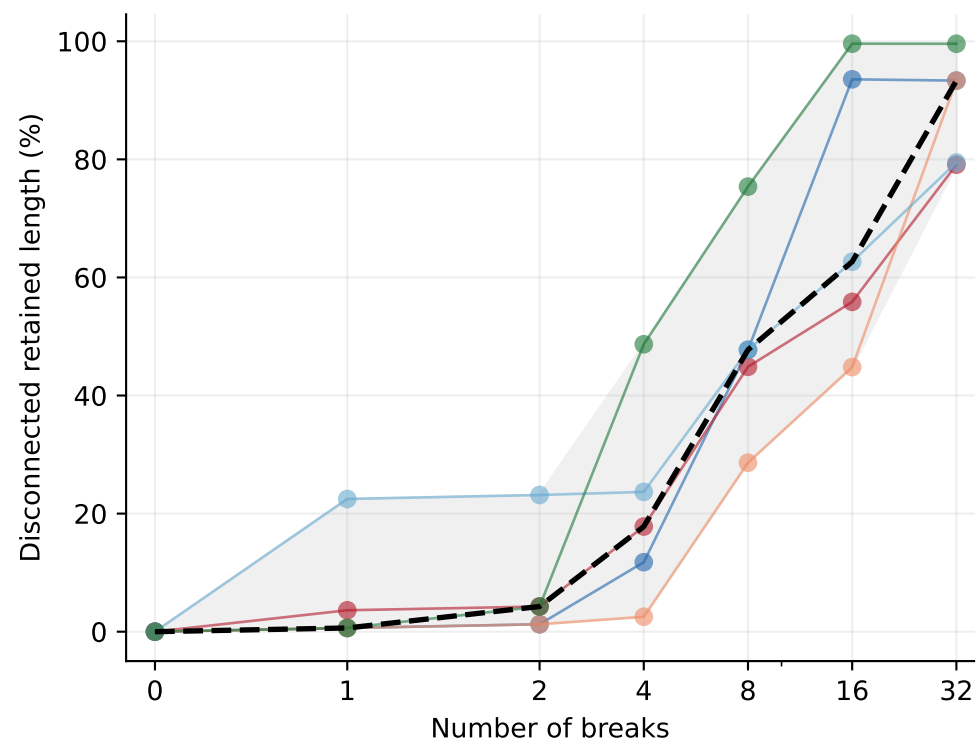
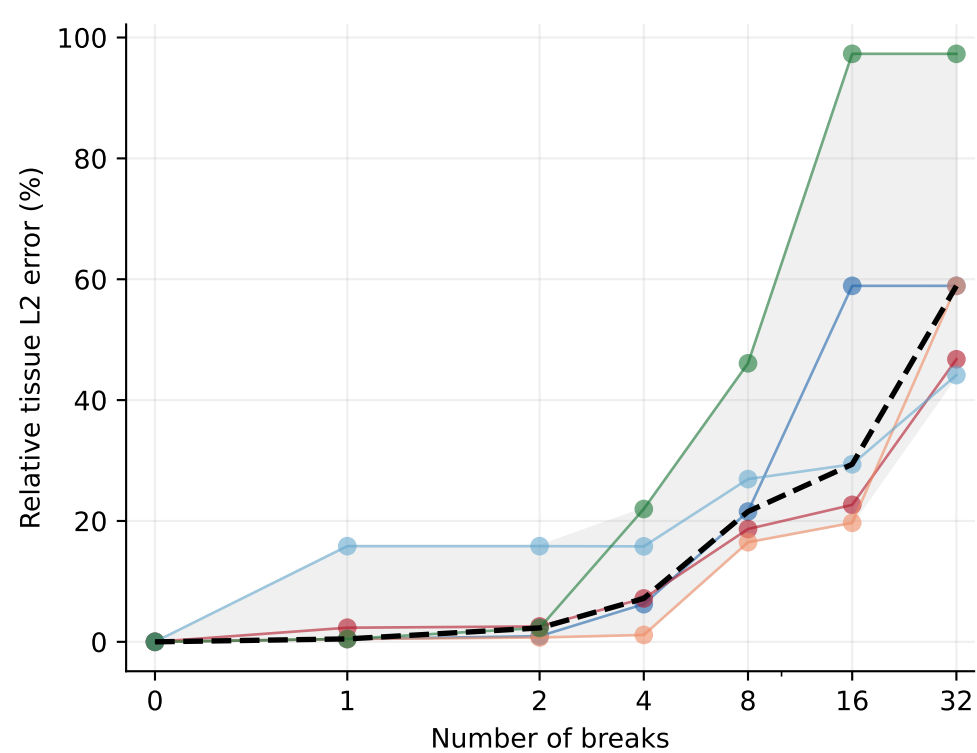
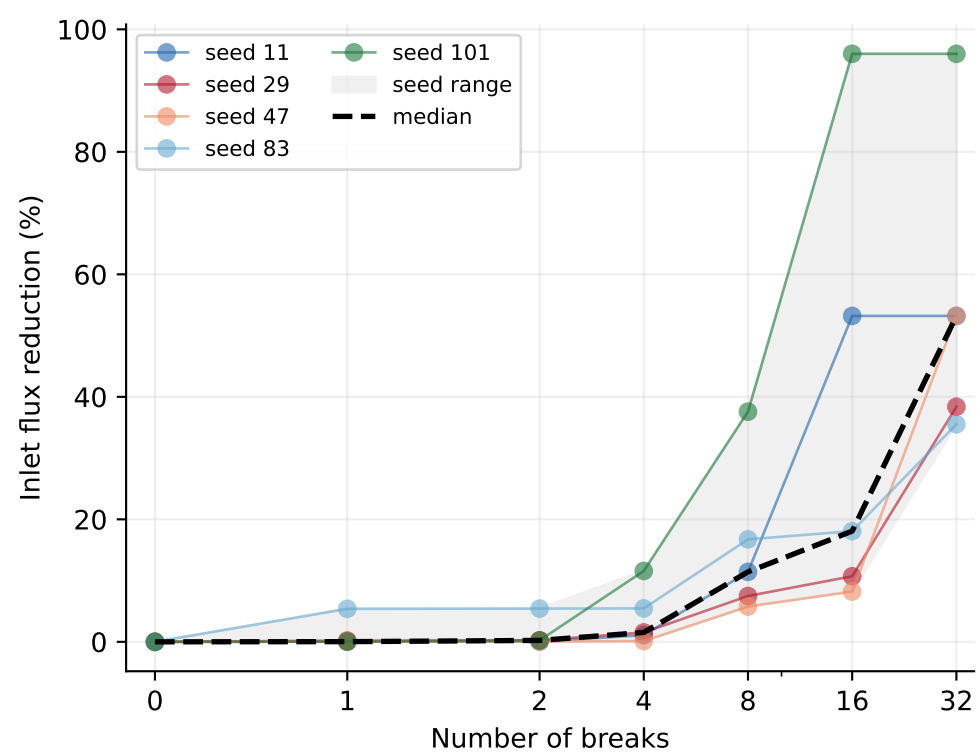
Y: global exchange response



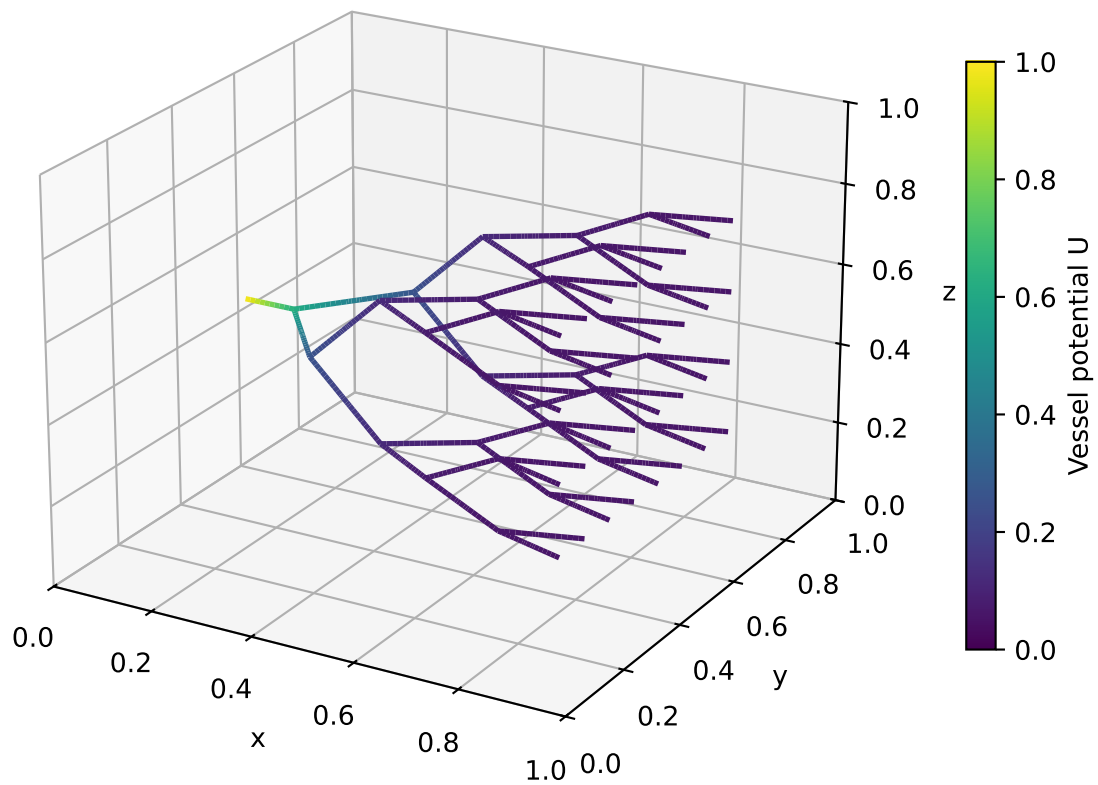
Y: change in tissue potential



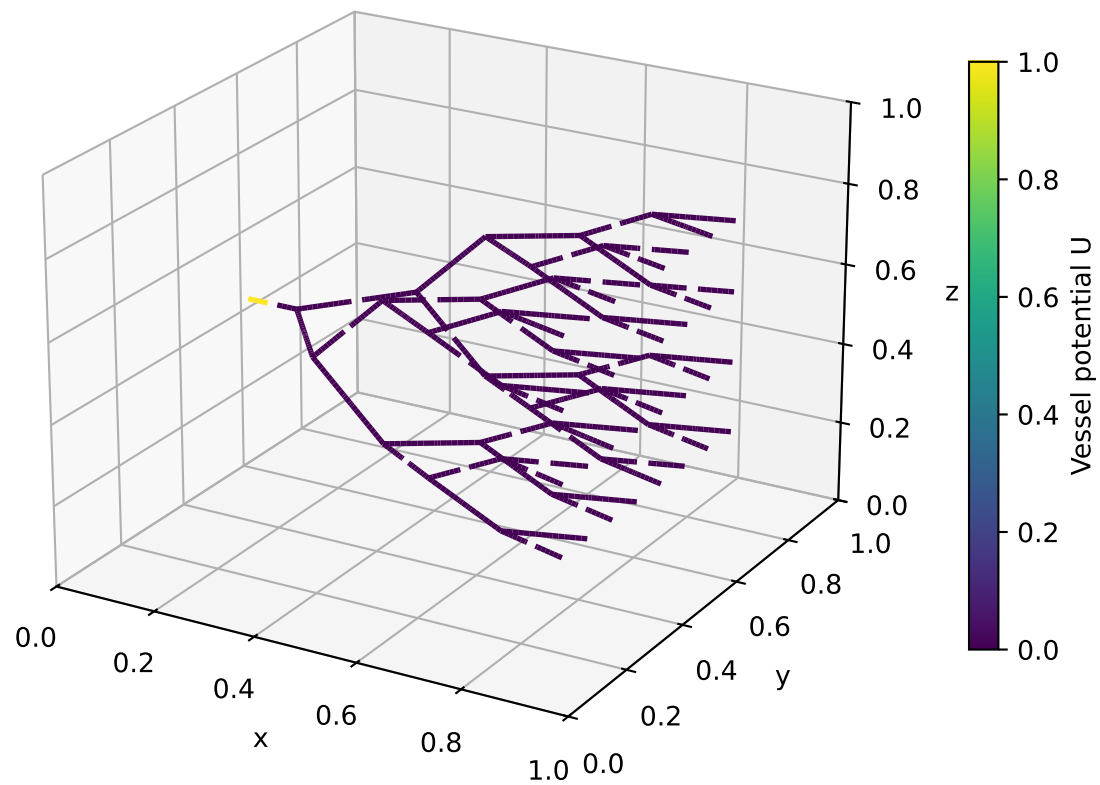
line: intact, $z=0.5$ line: broken, $z=0.5$ line: difference, $z=0.5$ y: intact, $z=0.5$ y: broken, $z=0.5$ y: difference, $z=0.5$ 



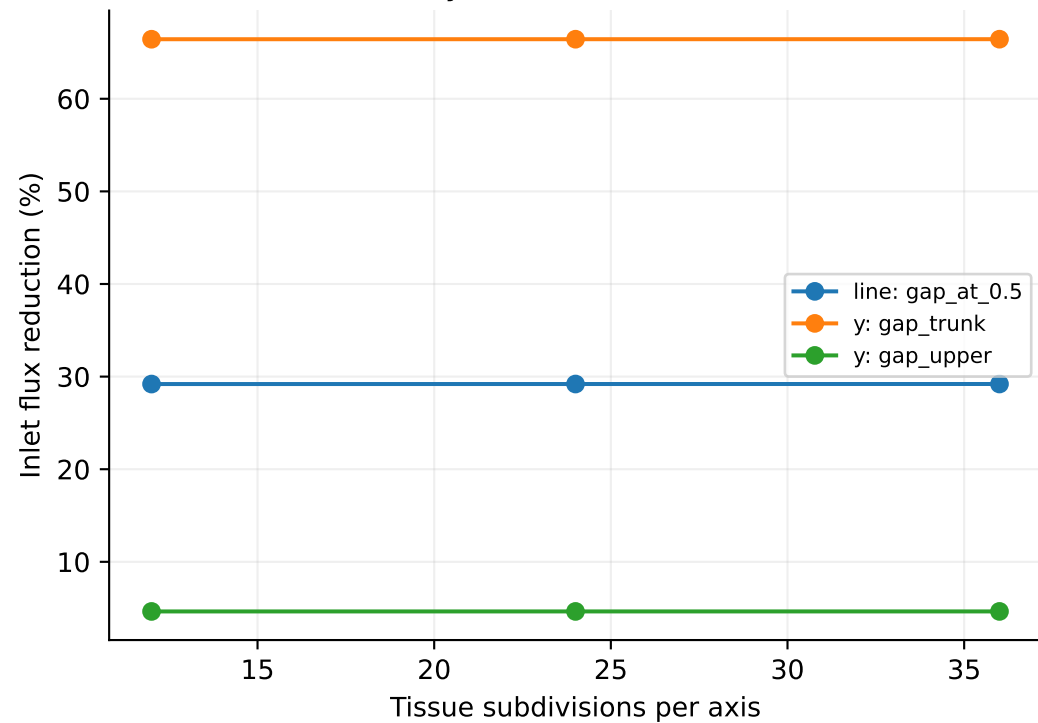
Intact tree



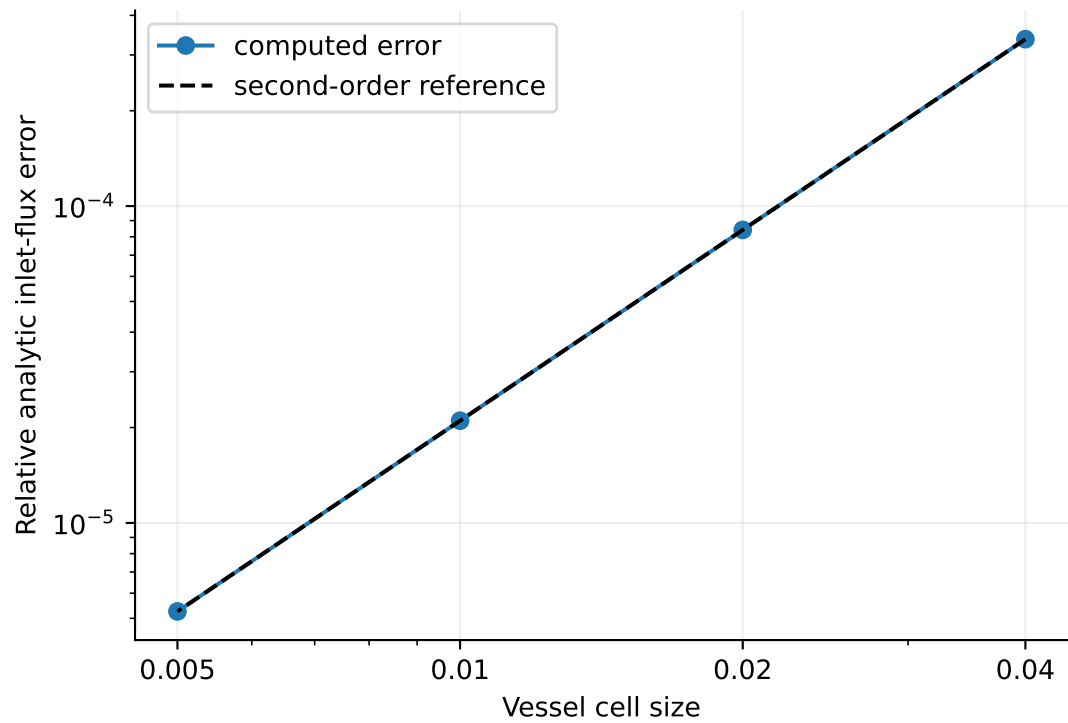
32 breaks, seed 101



Sensitivity versus tissue resolution



Closed-form limit check



Supplied-network simulation: pt002 inflow

Completed supplement, 11 September 2026. This run solves the intact inflow network from oilbloodata.zip, coupled to its supplied tissue mesh. A second solve checks the effect of halving the vessel cell size. Results are dimensionless.

Completed mesh files

- * coupled_solution.vtu: self-contained tissue and vessel mesh with both computed potentials and flux fields. This is the main deliverable and needs no companion file.
- * network_solution.vtu: vessel-only solution, convenient for a Tube filter in ParaView.
- * tissue_solution.vtu: full tissue solution.
- * metrics.csv: the main and refinement measurements.
- * verification.json: numerical and artifact checks.

The VTU coordinates are in the original millimetre coordinate frame. potential_dimensionless contains tissue u and vessel U . Cell field compartment_label is 1 for tissue and 2 for the inflow network; select label 2 with Threshold to expose the embedded network. The axial and tissue flux names end in dimensionless; wall exchange is per unit normalized arclength and is positive from vessel to tissue. Signed axial flow follows the source VTP edge orientation. Fields belonging to the other compartment contain zero placeholders. Source VTK point and edge IDs are zero-based; source_exodus_node_id preserves the one-based Exodus node IDs. Newly inserted vessel points have source point ID -1. Native XDMF/HDF5 files in the raw run directory retain normalized coordinates.

Data and inlet selection

The input files are oilbloodata/thermoembo_run/pt002/pt002_inflow_centerline.vtp and oilbloodata/thermoembo_run/pt002/mesh.exo. The archive README identifies the centerline coordinates as millimetres and the Exodus coordinates as metres. They were converted to one coordinate system before coupling.

The original network has 797 vertices, 828 edges, 55 endpoints, one connected component, and cycle rank 32. All edges and loops were retained, including 83 edges marked is_phantom in the source. Edge subdivision preserves their geometry and connectivity. All stored point and edge radii are 0.1 mm; this supplied value was retained rather than treated as a measured radius distribution.

The inlet is original point 0, at [66.1982421875, 99.6728515625, 475.5] mm. It is the unique degree-one vertex with the highest archived endpoint pressure, 836 mmHg. This supplies an explicit computational inlet selection. Archived pressures and flows come from a different resistance model and are used only to identify this endpoint; they are not calibration or validation targets for the new PDE solution.

Model and normalization

The model is the same Laurino-Zunino circular-average 3D-1D weak formulation used in the synthetic pilot:

$$\begin{aligned} & \text{Integral_Omega } K3 \text{ grad}(u) \cdot \text{grad}(v) \\ & + \text{Integral_Lambda } K1 A U_{\text{prime}} V_{\text{prime}} \\ & + \text{Integral_Lambda } \kappa P (U - T_r u)(V - T_r v) = 0. \end{aligned}$$

$K3=K1=1$, $\kappa=0.05$, $A=\pi r^2$, and $P=2\pi r$. $U=1$ at the selected inlet; other vessel endpoints have

natural zero axial flux. The entire exterior tissue boundary has $u=0$. Tissue and vessel potentials are continuous P1; circular averages use DG1 with quadrature degree 20. Source terms are zero. The circle checks include every endpoint interpolation location and additional midpoint samples.

Lengths are normalized by the longest extent of the supplied tissue mesh, $L=202.380554199$ mm, with origin $[-0.11389311981201172, 0.041460384368896486, 0.408371337890625]$ metres. Thus $r=0.0004941186192$ in the PDE. The imported tissue mesh is preserved: 126,569 tetrahedra and 27,947 vertices. Its original volume is 887.609130 mL. The main vessel discretization has 2,016 cells and 1,985 vertices, with maximum normalized cell length 0.005 (about 1.0119 mm). The coupled system has 29,932 pressure degrees of freedom and was solved on 4 MPI ranks using PETSc/MUMPS.

These are model potentials and dimensionless fluxes. The coefficients and inlet value have not been fitted to physiological measurements. The explicit use of the supplied radius and organ domain also means the numerical percentages from the earlier synthetic examples should not be transferred to this network.

Results

Quantity	Computed value
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Conservative inlet flux	1.23142857635e-05
Integrated vessel-to-tissue exchange	1.23142857635e-05
Tissue exterior outflow	1.23142857635e-05
Mean tissue potential (volume-normalized)	3.08670848106e-07
Maximum tissue nodal potential	2.95045856171e-05
Vessel nodal potential range	1.43228924415e-11 to 1
Assembly / linear solve time	115.91 s / 0.29 s

Validation

Check	Result
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Maximum relative flux imbalance	1.665e-14
Relative linear residual	1.234e-19
Relative energy-identity error	8.254e-16
Positive inlet value	1
Main coupling containment	66,528 samples; zero outside
Halved maximum 1D cell size	3,564 vessel cells; inlet-flux change 0.009984%
Existing toy-model regression	4.530e-14 maximum relative difference
Mesh preservation and VTU read-back	Passed

The refinement check concerns the one-dimensional discretization on the supplied tissue mesh. It does not establish three-dimensional mesh convergence. Input checksums and the coordinate transformation are in `supplied_input_metadata.json`; source IDs are embedded in the VTU files. Both runs' metrics are in `supplied_metrics.csv`. Unmodified extracted inputs and native XDMF/HDF5 solutions remain with the raw runs on NOTS.

Reproduce

From the study source directory on NOTS:

```
source src/environment.sh
```

```
python -m src.supplied --prepare
sbatch src/supplied.slurm
# After that job completes:
python -m src.supplied_results
python -m src.collect
python -m src.package
```

Reference: Laurino and Zunino (2019), equations (11)-(12).